

Scaling and PCA Report

Scaling of Numerical Features

All selected continuous acoustic features were scaled using **StandardScaler**.

Justification:

The extracted physical audio features operate on vastly different scales and physical measurement units. For example, frequency-based dimensions such as `spectral_centroid` reach values in the thousands of Hertz, whereas energy-based metrics like `rmse` and harmonic features like `tonnetz` are fractional decimals bounded below 1.0. If these features are analyzed in their raw form, variables with larger numerical magnitudes would artificially dominate variance calculations and distance metrics during analysis.

To prevent magnitude bias, **StandardScaler** was applied to standardize all 410 continuous features to a common scale with a mean of 0 and a standard deviation of 1. This ensures that every acoustic dimension—spanning timbre, loudness, harmonic pitch, and percussiveness—contributes equally during Principal Component Analysis.

Principal Component Analysis (PCA)

Purpose

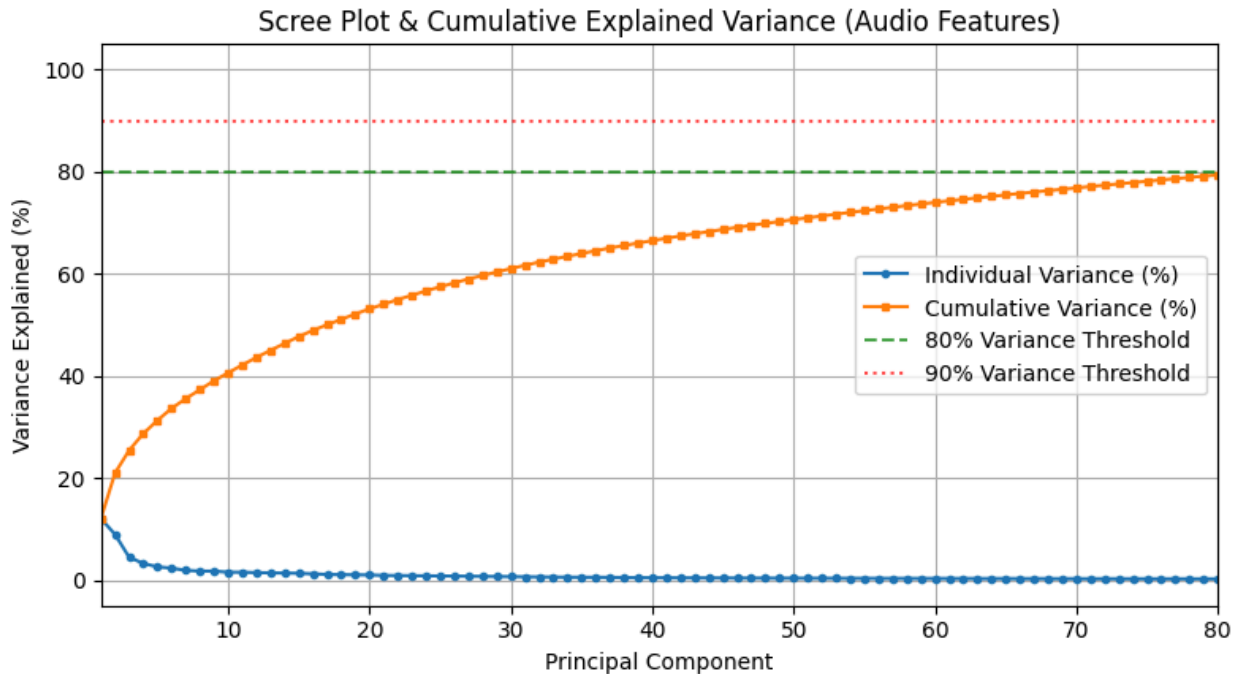
PCA was applied to evaluate the multi-dimensional structure of the 410 extracted audio features and identify which orthogonal axes explain the majority of variance across the dataset. It serves to condense high-dimensional acoustic data, eliminate multi-collinearity among overlapping statistical moments, and test whether commercial popularity tiers (high, medium, low) naturally separate in low-dimensional space.

Is PCA Necessary for This Dataset?

Strictly speaking, the raw acoustic extractors already possess direct physical and musical interpretations (such as brightness, harmonic contrast, and energy). Stakeholders can interpret individual audio dimensions directly in their domain context.

However, PCA is necessary in this project to:

- Address high multi-collinearity and redundancy across 410 continuous statistical moments (means, medians, standard deviations, skewness, and kurtosis).
- Determine the effective intrinsic dimensionality required to preserve 80% to 90% of total acoustic variance.
- Test the hypothesis of linear separability between hit and flop tracks in low-dimensional space. Support computational efficiency and prevent overfitting in downstream predictive modeling.



Results of PCA

- **Total features evaluated:** 410 continuous audio features
- **PC1 Explained Variance:** 12.05%
- **PC2 Explained Variance:** 8.94%
- **Cumulative Variance (PC1 + PC2):** 20.99%
- **Principal components selected ($\geq 80\%$ variance):** 83 components (79.8% dimensionality reduction)
- **Principal components selected ($\geq 90\%$ variance):** 141 components (65.6% dimensionality reduction)

The scree plot indicates a steep initial drop in individual explained variance across the first 10 components, reflecting the consolidation of shared collinear audio moments. Retaining 83 principal components is sufficient to preserve over 80% of the entire dataset's informational variance while eliminating 327 redundant dimensions.

Interpretation of Principal Components

- **PC1:** Represents spectral contrast and frequency harmonic separation (dominated by spectral_contrast mean and median sub-band moments).
- **PC2:** Represents spectral envelope shape, high-frequency brightness cutoff, and timbral sharpness (loaded by spectral_rolloff kurtosis and lower-order mfcc moments).
- **PC3:** Represents percussive noisiness and temporal energy dynamics (driven by zcr zero-crossing rates and spectral_centroid moments).
- **PC4:** Represents short-time Fourier pitch-class energy distribution (loaded by chroma_stft semitone dimensions).
- **PC5:** Represents Constant-Q harmonic pitch distribution and upper-frequency timbral nuances (driven by chroma_cqtkurtosis and higher mfcc coefficients).
- **PC6:** Represents base timbral texture versus tonal pitch minimum boundaries (loaded by base mfcc moments and chroma_cqt minimums).
- **PC7:** Represents spectral frequency spread and dynamic bandwidth variability (driven by spectral_centroid, spectral_rolloff, and spectral_bandwidth standard deviations).
- **PC8:** Represents mid-frequency timbral variance and stability (dominated by mid-range mfcc standard deviations).
- **PC9:** Represents specific semitone frequency concentrations (loaded by higher-order chroma_cqt and chroma_stftmoments).
- **PC10:** Represents tonal space centroid movement and fifths/thirds harmonic pitch relationships (driven by tonnetzmean and median dimensions).

Justification of PCA Usage

PCA was deployed to reduce feature redundancy and uncover global structural patterns within the high-dimensional acoustic feature space. Signal processing extractors naturally produce correlated statistical moments across frequency, pitch, and timbre.

Key justifications for using PCA in this analysis include:

- **Dimensionality Compression:** Reduces the feature space from 410 variables to 83 principal components, achieving a 79.8% reduction in model input size while retaining 80% of total variance.
- **Evaluating Linear Separability:** Projecting tracks onto the primary components (PC1 vs. PC2) demonstrates that commercial popularity tiers form a continuous, overlapping distribution rather than isolated linear clusters.
- **Guiding Algorithm Selection:** The absence of discrete linear clusters in PCA space empirically justifies that simple linear models are insufficient, supporting the selection of

non-linear ensemble models (such as Random Forests and Gradient Boosting) for classification.

While individual physical sound features remain valuable for direct feature-level interpretation, PCA provides an objective mathematical framework to compress the feature space and eliminate multi-collinearity for machine learning readiness.

Conclusion

In this stage, the continuous acoustic dataset was standardized and evaluated through Principal Component Analysis.

All 410 continuous numerical audio features were scaled using StandardScaler to ensure that features with large physical units (such as Hertz) did not disproportionately dominate the analysis over decimal-scale energy features. Full PCA decomposition revealed that 83 principal components capture over 80% of total variance, successfully reducing the dimensional complexity of the dataset by 79.8%.

Furthermore, 2D principal component projections revealed a continuous, overlapping distribution across popularity classes, proving that track commercial performance is governed by non-linear acoustic interactions rather than simple linear boundaries. This analysis confirms that the feature space is preprocessed, non-collinear, and ready for non-linear predictive modeling.

